ChatGPT

Simulation and Reinforcement Learning for Autonomous Robotics with Isaac Sim and ROS 2

(Robotic Cloth Manipulation via the MDP Workflow of IsaacLab, IsaacSim 5.1)

1. Introduction *(~5 pages)*

- **Motivation and Context:** Introduce the importance of autonomous robotic cloth manipulation in manufacturing and recycling contexts. Highlight the challenges of handling deformable objects (textiles) and why simulation with reinforcement learning (RL) is a promising approach. Emphasize the specific scenario: using a 6-DOF tendon-driven robotic arm to pick up and spread cloth for inspection on a conveyor, as outlined in the project's problem statement [1] [2].
- **Problem Statement & Objectives:** Clearly state the thesis problem: developing an RL-based control strategy for a tendon-driven robot to manipulate cloth (grasp, stretch, present) in simulation and eventually real life [1]. List the research objectives, for example: (1) establishing a high-fidelity Isaac Sim simulation environment, (2) designing and training RL policies for cloth manipulation tasks, (3) integrating a custom tendon-driven robot model, and (4) addressing sim-to-real transfer (ensuring the learned policy can generalize to real-world or be adapted) [2]. Explain that initial simpler manipulation tasks have already been implemented as a stepping stone (e.g. basic pick-and-place with a generic robot), and the thesis will build on this foundation toward the full cloth manipulation scenario.
- **Thesis Outline:** Summarize the structure of the thesis (chapters 2–7) for the reader. For instance: Chapter 2 provides background theory and related work; Chapter 3 identifies the research gap and refines the problem; Chapter 4 details the methodology including the simulation setup and RL approach; Chapter 5 presents experimental results; Chapter 6 discusses the findings and limitations; Chapter 7 concludes the thesis and outlines future work. This guide ensures clarity and aligns with FAPS guidelines for academic theses structure.

2. Background and State of the Art *(~20 pages)*

(This chapter thoroughly reviews fundamental concepts and prior work, providing the theoretical and technical foundation for the thesis.)
- **2.1 Reinforcement Learning for Robotic Manipulation:** Introduce Markov Decision Processes (MDP) as the formalism for RL, and outline key RL algorithms relevant to continuous control of robots. Cover model-free policy gradient methods (e.g. DDPG, PPO, SAC) commonly used for manipulation tasks, and discuss why these algorithms are suitable for high-dimensional, continuous action problems like controlling robotic arms. Highlight any specific algorithms or frameworks used in this thesis (e.g. Stable Baselines or NVIDIA's RL Games) and prior successes of RL in simulation for manipulation tasks [3].
- **2.2 Sim-to-Real Transfer Principles:** Explain the challenge of the "reality gap" between simulation and physical robots, and why sim-to-real transfer is critical for deploying learned policies. Discuss techniques such as domain randomization and domain adaptation – e.g. randomizing visual and physics parameters in simulation so that the real world appears as just another variation [4]. Mention successful examples of sim-

to-real in robotics (for perception and control) and how the Isaac Sim/IsaacLab platform is designed to facilitate sim-to-real (with photorealistic rendering and accurate physics [5]).
- **2.3 Cloth Simulation in Robotics:** Provide an overview of how deformable objects like cloth are modeled and simulated. Describe common modeling approaches (e.g. mass-spring models, particle systems, finite element methods) and the challenges of simulating cloth dynamics (high degrees of freedom, self-collision, computational load). Emphasize the unique challenges cloth poses: the deformable nature of textiles breaks standard robotic assumptions (rigid bodies, simple dynamics) – applying forces to cloth not only moves it but also alters its shape [6]. Summarize current capabilities in simulation engines: for instance, Isaac Sim 5.1 includes support for deformable object simulation (cloth) with NVIDIA PhysX, which is crucial for creating realistic training scenarios [7]. Also review any relevant physics parameters (stretching, bending stiffness) and how they might be tuned for realism.
- **2.4 NVIDIA Isaac Sim and IsaacLab Architecture:** Introduce NVIDIA Isaac Sim (version 5.1) as the simulation environment used. Describe its core features – built on Omniverse, providing high-fidelity physics (PhysX 5) and photorealistic rendering with RTX, support for ROS 2, and GPU acceleration for parallel simulations [8]. Explain IsaacLab, NVIDIA's unified robotics learning framework built on Isaac Sim (evolved from the Orbit framework) [9]. Note that IsaacLab combines Isaac Gym's GPU-accelerated physics with Orbit's environment library, providing a modular toolkit for RL, imitation, and motion planning [9]. Discuss how IsaacLab's *MDP workflow* organizes simulation tasks: defining robots, environments, and training pipelines in a reusable way. This subsection should make clear why IsaacLab + Isaac Sim 5.1 is well-suited for this thesis (e.g. integrated cloth dynamics support for realistic learning [7] and tools for distributed RL training).
- **2.5 Tendon-Driven Robot Kinematics and Modeling:** Explain what tendon-driven robots are – actuators apply force via tendons (cables) to move joints or continuum segments, often allowing lighter, more compliant designs but requiring complex modeling. Cover basic kinematic principles: e.g. how pulling tendons causes joint rotation or continuous bending, constant-curvature assumptions in continuum segments, and how inverse kinematics/control differ from rigid joint robots. Mention any specific robot model relevant to the project (if a particular tendon-driven arm design is given). Discuss the challenges in simulating tendon-driven systems in an environment like Isaac Sim – such as modeling the tendons' routing and elasticity, which might be approximated via constraint forces or via a URDF/MJCF model with mimic joints, etc. Include any literature on tendon-driven manipulators for context, e.g. how their precision and dynamics differ from direct-drive robots.
- **2.6 Policy Learning Pipeline (MDP Workflow):** Detail the standard reinforcement learning pipeline in the context of IsaacLab's MDP workflow. Define the components of the MDP for the cloth manipulation task: state (observations), action space (robot joint torques/velocities or high-level gripper motions), reward function design (what constitutes success for grasping/stretching cloth), and episode termination conditions. Explain the concept of curriculum learning or shaping rewards if used to help the policy learn complex tasks in stages. This subsection essentially bridges theory to implementation, describing how a manipulation task is formulated for an RL agent. If IsaacLab provides any specific API or structure for tasks (such as environment wrappers, vectorized environments on GPU, etc.), mention how those are utilized [10].
- **2.7 Prior Work in Robotic Cloth Manipulation:** Review the existing research on robotic cloth or deformable object manipulation. Cover both model-free approaches (e.g. deep RL or learning-based methods) and model-based or analytical approaches (e.g. vision-based controllers, heuristic methods). Cite a few key studies or benchmarks – for example, folding and unfolding tasks, cloth smoothing/flattening (like work by Seita et al. on fabric smoothing, or the GarmentBench/GarmentLab benchmarks in recent literature). Note the outcomes: e.g. model-free RL has been applied to fabric smoothing and achieved good performance in simulation, but often needs many training samples and can struggle to generalize beyond the trained task [11] [12]. Highlight open problems identified in surveys: generalizing to varying fabrics, dealing with noisy perception, and accomplishing long-horizon multi-step cloth tasks are still challenging

[13] [14]. This subsection situates the thesis in context – how it builds on prior work (e.g. using RL in sim for cloth) and what gaps it aims to address (perhaps combining tendon-driven hardware with cloth manipulation, or improving sim-to-real for deformables).
- **2.8 Robot and Scene Description Formats (USD, URDF, MJCF):** Briefly explain the file/format standards used in this project for simulation models. USD (Universal Scene Description) is used for constructing complex simulation scenes in Isaac Sim [15] – describe it as an extensible 3D scene graph format that can include objects, lighting, physics properties, etc. URDF (Unified Robot Description Format) and MJCF (MuJoCo XML) are common robot model formats; clarify that the tendon-driven robot is modeled in one of these (URDF or MJCF) for import into Isaac Sim [15]. Mention why these formats are needed: URDF/MJCF capture the robot's kinematic structure, joint limits, and inertial properties, while USD integrates the robot and cloth into the simulated world. Ensuring consistency between these formats is important (e.g. using the URDF to define the robot, which Isaac Sim can import, and using USD for the overall environment configuration) [15]. This section may also note any modifications done to the robot's URDF to model tendons (since standard URDF may not natively support tendon constraints, workarounds or custom plugins might be used).

(By the end of the Background chapter, the reader should understand all key concepts – RL, sim-to-real, cloth physics, the IsaacSim/IsaacLab toolchain, tendon-driven robot modeling – and see how they interrelate in the context of this thesis. This comprehensive background will be about one-third of the thesis, aligning with academic standards.)

3. Problem Statement and Research Gap *(~3 pages)*

(This chapter identifies the gap in existing knowledge and defines the specific research questions that the thesis addresses, following from the background review.)
- **Critical Analysis of State of the Art:** Build directly on Chapter 2's review – summarize what has been done in cloth manipulation and where limitations remain. For example, note that while prior works have achieved either cloth manipulation with traditional robots or basic RL policies in simulation, the combination of a novel tendon-driven arm and complex cloth tasks presents open challenges. Point out unresolved questions (e.g. "How to effectively train an RL policy for a high-DOF tendon-driven manipulator on cloth tasks without extensive manual reward shaping?", or "How to maintain sim-to-real fidelity for cloth interactions and tendon dynamics?"). This section should **critically discuss** the literature: what are the *deficits or gaps* in current approaches that justify the thesis work [16] [17].
- **Research Gap and Justification:** Clearly articulate the *Handlungsbedarf* ("need for action") as per FAPS guidelines [17]. For instance: *"Most existing studies on cloth manipulation use rigid-joint robots or simplified simulations; the literature lacks demonstrations of RL-based control on tendon-driven robots for cloth handling. Moreover, sim-to-real strategies for such deformable tasks are not well established."* Justify that addressing this gap has both scientific value (contributing knowledge on tendon-driven robot control and RL in deformable object tasks) and practical relevance (e.g. automating textile recycling or clothing handling tasks) [18].
- **Refined Problem Definition:** Based on the gap, formulate the precise research questions or hypotheses of this thesis. This might include: "How can a simulation environment be configured to realistically train cloth manipulation policies for a tendon-driven robot?", "Which RL algorithms and reward formulations yield stable learning for the cloth stretching task?", and "What strategies can improve the transferability of the learned policy to a physical robot (using ROS 2)?" Also define the *scope* of the project – e.g. focusing on a specific cloth size/type, a specific robot design, and what is *not* addressed (such as vision-based perception might be fixed or simplified, etc., if applicable). If appropriate, list the main contributions of the thesis in bullet points (e.g. *1) integration of a tendon-driven robot model in Isaac Sim, 2) an RL pipeline for cloth grasp-*

and-stretch, 3) evaluation of policy performance and generalization).
- **Success Criteria:** (Optional) Define how success is measured. For example: achieving a certain cloth spreading coverage in simulation, stability of the tendon arm control, and comparisons to baseline methods. This lets the reader know what to look for in results, and ties the problem statement to the forthcoming methodology.

(By clearly formulating the problem and gap, this chapter ensures the reader knows exactly what questions the thesis will answer. It sets the stage for the approach described next. In line with FAPS standards, this acts as a bridge from theory to the method, derived from the literature review [19] [20] *.)*

4. Methodology and Implementation *(~15–18 pages)*

(This chapter describes how the research was conducted and the system developed. It should be detailed enough for reproducibility, covering the simulation setup, robot and environment configuration, RL algorithm, and experiment design. According to FAPS guidelines, together with results and discussion, this methodology forms the core of the thesis, about two-thirds of the content [21] *.)*
- **4.1 Overview of Approach:** Provide a high-level overview of the solution strategy. Possibly include a diagram of the system architecture or workflow (showing the IsaacLab environment loop: state → policy → action → environment). Describe how the development progressed in stages: for example, *Stage 1:* set up a simple simulation with a generic robot and train basic RL tasks; *Stage 2:* integrate the tendon-driven robot model; *Stage 3:* tackle the cloth manipulation task with the full system. This overview helps orient the reader before diving into details.
- **4.2 Simulation Environment Setup:** Detail the configuration of the Isaac Sim 5.1 environment used for training. Describe the scene constructed in USD: the table or conveyor, the cloth object (size, mesh or primitive used, physical properties like mass, friction, stiffness), and the initial conditions (e.g. cloth crumpled or folded in a consistent way for each episode, or randomized drops). Include the robot's placement and any sensors in simulation. If ROS 2 is utilized within Isaac Sim (through the ROS bridge extension), explain how ROS topics are configured (for instance, to publish observations or accept action commands). Also mention simulation parameters: physics time-step, whether the simulation is run in faster-than-real-time, etc., to facilitate RL training.
- **4.3 Robot Model Integration (Tendon-Driven Arm):** Explain how the 6-DOF tendon-driven robotic arm is modeled and imported. For example, if using URDF: describe how joints are represented and how tendon coupling is modeled (URDF might not directly model tendons, so perhaps joints are connected via mimic joints or a plugin). If using MJCF, mention that MuJoCo's format can model tendon wraps and was used, then converted or loaded in Isaac Sim. Discuss any verification of the robot model in simulation – e.g., testing that moving the joints via torque/position control results in expected motion, calibrating the tendon tensions or spring constants so that the simulation reflects the real robot's behavior. If the robot's physical parameters (masses, link lengths) came from a CAD or prior data, note that. Essentially, show that the simulated robot is a valid representation of the real tendon-driven arm (perhaps including a figure of the robot model). This section addresses the "Validate physicality of model" step from the project timeline [22] [23] .
- **4.4 Reinforcement Learning Setup:** Describe in depth how the RL problem is formulated. This should cover:
- **State Space:** What observations the agent receives. For instance, the state might include the robot's joint angles/velocities, gripper status, and possibly features of the cloth (if using explicit state like key points or shape descriptors). If vision (camera images) is used, mention how images are processed or if a learned encoder is part of the policy. Otherwise, maybe the cloth state is partially observed or represented by hand-

crafted features (e.g. corner positions if known).
- **Action Space:** What actions the agent outputs. Likely continuous joint commands (position increments or velocity or torque) for each of the robot's DOFs. Note if actions are low-level (joint torques via IsaacGym/PhysX) or higher-level (position increments or end-effector target poses). If a tendon-driven arm has any special actuation (like multiple joints moving together via one tendon), clarify how the action space controls those (maybe directly controlling joint angles which indirectly tension tendons).
- **Reward Function:** Provide the reward design for the cloth manipulation task. For example, a reward could be based on how flat or stretched the cloth is – perhaps negative distance of cloth state from a desired flat configuration, or increase in covered area on the table [24]. Include intermediate shaping rewards if used (e.g. a reward for grasping the cloth successfully, another for moving it upward, etc., to guide the agent). If the task is staged (first learn to grasp, then to stretch), describe the reward schedule or curriculum. Emphasize that reward design is critical and any iterative tuning that was done.
- **RL Algorithm and Training Parameters:** State which RL algorithm is used (e.g. Proximal Policy Optimization given its stability for continuous actions). Note the library (IsaacLab likely integrates with RL frameworks like RL Games or Stable Baselines [25]). Provide key hyperparameters: learning rate, discount factor, policy network architecture (how many layers/neurons, any LSTM if needed), training duration (number of simulation steps or episodes). If training was distributed (multiple parallel environments on GPU), mention how many parallel simulations were used to speed up learning (IsaacLab can leverage GPU acceleration for many parallel instances [10]). Also mention if any pre-training or demonstrations were used, or if it was pure reinforcement learning from scratch.
- **MDP Workflow in IsaacLab:** Tie the above into IsaacLab's structure – e.g., "Using IsaacLab, we defined a custom task class for the cloth manipulation scenario, specifying its observation and reward functions as above, and we utilized IsaacLab's wrappers to interface with the RL algorithm." If ROS 2 plays a role in training (for instance, maybe observations or actions go through ROS messages), describe that integration; however, typically training might be internal, and ROS 2 would be used later for connecting to real hardware.
- **4.5 Initial Experiments: Basic Task Policies:** Describe the initial RL experiments conducted with simpler tasks and robots (as mentioned, the student has *already implemented initial RL policies in Isaac Sim for basic tasks*). For example, this could be a **pick-and-place task with a simple rigid robot** (perhaps a standard manipulator) to validate the RL pipeline. Detail one such task: e.g., picking up a rigid object or a corner of a cloth stand-in. Explain the purpose: to test the training setup, reward shaping, etc., before introducing the complexities of the tendon robot and deformable cloth. Summarize the setup differences: a simpler robot model (maybe a 6-DOF arm with direct joint control), possibly a dummy object instead of a flexible cloth. Note any outcomes (briefly, as results will be in next chapter) but focus on methodology: what was learned or adjusted from these initial trials (e.g., needed to tune hyperparameters, or found that a certain observation was missing and then added). This acts as a proof-of-concept phase.
- **4.6 Advanced Task Implementation – Cloth Grasping and Stretching:** Detail the final task scenario which the thesis addresses: using the tendon-driven robot to grasp a piece of cloth from a surface and stretch it out. Break down how this complex task is implemented: for instance, the task might be divided into phases (grasp, lift, stretch, place). Describe any specific strategies used to make learning feasible: such as curriculum learning (start with cloth already grasped, or start with easier initial cloth configurations and progressively increase difficulty), or auxiliary rewards for sub-goals (like a reward for just successfully grasping the cloth, to encourage the first step). If the action space or policy needed to be extended (for example, controlling a gripper to actually pinch the cloth – mention if a gripper is simulated and how grasp detection is handled, possibly via Isaac Sim's contact sensors or a constraint when the gripper closes on cloth). Also note any safety or stability measures (like limiting the force on tendons to realistic ranges to prevent simulation explosions). This subsection essentially describes how the *target task* is set up in simulation in its full complexity.

- **4.7 Software and Hardware Integration (ROS 2):** Finally, describe how ROS 2 is incorporated, especially with an eye towards real-world deployment. If during simulation experiments, ROS 2 was mainly used as a bridge (e.g., echoing states or visualizing in RViz), mention that. More importantly, outline how the trained policy could interface with a real robot via ROS 2: for instance, the robot's joint controllers can subscribe to commanded positions from the RL policy, and the policy can subscribe to sensor feedback (joint states, possibly a depth camera for cloth observation if going real). In the context of this thesis (which is primarily simulation-focused), this section may discuss any preliminary setup for sim-to-real, such as using ROS 2 control for the tendon robot in sim to mirror what would be done on hardware, or exporting the trained policy for a ROS 2 node. It sets the stage for future work but also confirms the system is designed with real-world integration in mind.

(Through the Methodology chapter, the reader should see exactly how the research is executed. Each step – from setting up the simulator to training the RL agent – is documented. This level of detail ensures reproducibility and aligns with guidelines (methods must be transparent and allow others to replicate the study [26] [27]). Roughly 15–18 pages are allocated here, which combined with results and discussion meets the recommendation that the core chapters form about 2/3 of the thesis [21].)

5. Results *(~15 pages)*

(This chapter presents the outcomes of the experiments described in the methodology. It should be organized logically, often mirroring the methodology subsections. The results are presented factually and without interpretation – save analysis for the Discussion chapter, although brief commentary for clarity is fine. Include figures like training curves, success rate plots, and illustrative images of the robot's actions. Ensure the data presented ties back to the objectives and research questions.)

- **5.1 Preliminary Experiments (Baseline Results):** Start with results from the initial basic tasks (as implemented in section 4.5). For example, report the performance of the RL policy on the simple pick-and-place or reaching task with the simplified robot. This could include: learning curves showing reward over training iterations for the baseline task, the number of episodes needed to reach a success criterion, etc. If multiple algorithms or settings were tried in this phase, compare them briefly (e.g. "PPO vs DDPG in the reaching task"). The purpose is to demonstrate that the RL pipeline works on a simpler problem and to record any insights (perhaps the policy achieved near 100% success in simulation on the easy task after X timesteps, etc.). This section sets a reference point and possibly justifies choices for the harder task (for instance, confirming the hyperparameters or algorithm).
- **5.2 Performance of Tendon-Driven Robot on Simple Tasks:** If the tendon-driven arm was also tested on a simple task (for example, after integration, maybe the same pick-place was attempted with the new robot to validate its controllability), present those results. This might include any differences observed due to the robot's dynamics – e.g., learning might have been slower or the policy had to deal with more complex dynamics. Show any quantitative metrics like success rate or final accuracy for the robot achieving the basic manipulation. Additionally, any validation of the simulated model could be reported here (e.g. "the tendon-driven robot's joint motions in simulation corresponded well to expected behavior; an open-loop test of moving one joint resulted in the correct end-effector displacement, confirming model accuracy").
- **5.3 Cloth Manipulation Task Results:** This is the main results section. Present the outcomes of training the RL agent on the cloth grasping and stretching task. This will likely include:
- **Learning Curve:** A plot of reward vs training steps, or success rate vs episodes. Discuss how quickly (or slowly) the agent learned to make progress, and whether it plateaued at a certain performance.
- **Quantitative Performance:** Define a success metric for the cloth task (e.g. percentage of cloth spread out flat above a threshold). Report the final performance of the policy according to this metric, perhaps

averaged over many test episodes. For example, "the policy achieves on average 85% cloth coverage after manipulation, compared to 20% initial, with a success (>=90% coverage) in 70% of trials." Use tables or graphs as needed to summarize these.
- **Policy Behaviors:** Describe what the learned policy actually does. For instance, "The agent learned to grasp a corner of the cloth near an edge, then lift and shake it before laying it down stretched." This can be illustrated with sequential images from simulation (embedded if allowed) showing keyframes of the robot and cloth. If any interesting behaviors emerged (like the robot re-grasping the cloth, or pulling in a particular direction), note them.
- **Ablation or Variant Results:** If additional experiments were run (e.g., trying different reward formulations, or disabling certain observations to test their importance), present those results briefly. For example, "We tried a variant without tactile feedback – success rate dropped by 15%, indicating the value of contact sensors." Keep this focused on results rather than interpretation.
- **Sim-to-Real Considerations:** If any tests were done to gauge sim-to-real readiness (perhaps running the policy with added noise, or a preliminary test on a real robot if available), include those results. For instance, "When moderate observation noise was added to simulate sensor noise, the policy performance degraded only slightly, suggesting some robustness." If the real hardware was partially involved (even if not fully within scope), mention any outcomes (e.g., "Initial trials on the physical robot with the learned policy showed it could execute the motions, but grasping the real cloth was inconsistent – see Discussion").
- **5.4 Summary of Key Results:** (Optional sub-section) End the results chapter with a concise recap of the most important findings. Bullet-point the answers to the objectives as evidenced by results, e.g.: *Achieved reliable cloth picking in simulation; Partial stretching accomplished (~85% flatness); RL training converged in ~N million steps; etc.* This prepares the reader for the next chapter, where these results will be discussed in context.

(All figures and tables in this chapter should be properly labeled and referenced in the text. The tone remains neutral, "reporting" what was observed [28]. Interpretation words like "because" or comparisons to expectations are saved for Discussion. By dedicating ~15 pages, there is room for multiple graphs and images which is often necessary for a robotics thesis. The structure mirrors the methodology to make it easy to connect each result to the corresponding part of the approach.)

6. Discussion *(~5 pages)*

(In this chapter, you interpret the results, discuss their implications, and relate them back to the research questions and prior work. This is where you explain what the results mean and assess the success of the project. FAPS guidelines emphasize critical discussion of results in light of the initial problem [29] [30].)
- **6.1 Evaluation of Outcomes:** Discuss to what extent the thesis objectives (from Introduction) were achieved. For example: Was a robust policy learned for the cloth manipulation task? How effective is it, based on the results? Address each objective: if the goal was to stretch the cloth for inspection, did the robot manage to present the cloth flat enough consistently? Provide reasoning for the performance observed – e.g., "The learning curve suggests the agent plateaued; this could indicate the policy found a suboptimal but acceptable strategy (like only partially stretching the cloth) and further improvement was difficult due to the sparse reward beyond that point." Connect this with theory: perhaps the high-dimensional state of the cloth made it hard to get a dense reward, or the exploration was challenging.
- **6.2 Comparison with Related Work:** Compare your approaches and results with those in literature (from section 2.7). For instance, if prior work on cloth folding with RL achieved a certain success or had to use demonstrations, discuss how your approach fared without demos, or how the complexity of using a tendon-driven robot compares to others using standard robots. If your results are better in some sense (maybe

using GPU simulation allowed more training and thus a better policy), point that out. Conversely, analyze any shortcomings: e.g., *"Unlike the fabric smoothing by Seita et al. [11] which reached near-perfect smoothing on real robots via imitation learning, our pure RL approach, while effective in simulation, might require additional strategies for real-world deployment."* This situates the contribution of the thesis: either showing improvement or highlighting differences.
- **6.3 Limitations of the Study:** A candid look at what was not achieved or what issues arose. Discuss the limitations in simulation fidelity – for example, the cloth simulation might still be an approximation (PhysX may not capture all fine wrinkles), and the tendon model might not include cable elasticity perfectly. Also, note if the policy might be overfitting to the simulated environment quirks (e.g., relying on a deterministic physics outcome). Mention any generalization concerns: does the policy only work for a specific cloth size or initial pose? What about varying the material properties? Additionally, if ROS 2/hardware integration was not fully realized, state that the results are currently limited to simulation. Acknowledge if training time was very high, or if the policy is sensitive to parameter changes. This honesty is important academically and will feed into future work.
- **6.4 Practical Implications:** Discuss what the results mean for real-world applications. For example, *"The success in simulation is a promising step toward automated cloth handling in recycling facilities. However, the gap to reality (sensor noise, unpredictable cloth variations) needs to be addressed before deployment."* If the tendon-driven robot in simulation behaved similarly to what you expect in reality, that's positive for transfer – mention that. If not, explain what adjustments might be needed (maybe more robust controllers at low level, or training with more randomness). This section can also reflect on the decision to use IsaacLab/ Isaac Sim: was it effective? Did GPU acceleration meaningfully speed up training (thus making such research feasible)? These insights can guide other researchers or engineers.
- **6.5 Recommendations and Lessons Learned:** Summarize any key lessons from the development process. For instance, *"We found that careful reward shaping was essential; a purely sparse reward for cloth stretching led to no learning. Introducing incremental rewards at different stages of the task was crucial."* Or *"When integrating the tendon model, we learned that calibration of the joint limits and stiffness in simulation greatly affects the training stability – an improperly tuned model caused unstable training."* These reflective points show critical thinking and are valuable takeaways beyond just the raw results.

(The discussion should be a thoughtful narrative, not just a bullet list of issues. It ties the thesis together by linking back to introduction and background, and showing how the work contributes to knowledge. Clarity here demonstrates the student's deep understanding of the project's outcomes. Roughly 5 pages is sufficient for a detailed discussion at this level of complexity.)

7. Conclusion and Outlook *(~3 pages)*

(The final chapter concludes the thesis, summarizing achievements and suggesting future work. FAPS guidelines note that conclusion + outlook should be succinct (not more than 3 pages) [31] [32] .)
- **7.1 Conclusion – Summary of Findings:** Recap the core accomplishments of the thesis in a few concise paragraphs. Reiterate the problem addressed and state how it was solved: e.g., *"In this thesis, we developed a reinforcement learning framework for a tendon-driven robotic arm to manipulate cloth in simulation. We successfully built a high-fidelity Isaac Sim environment with cloth physics and integrated a custom tendon-driven robot model. Using policy-gradient RL, the robot learned to pick up and stretch a cloth, achieving an X% improvement in coverage on average."* Highlight the significance: this might be one of the first demonstrations of RL on a tendon-driven continuum robot for cloth manipulation, or a novel combination of IsaacLab with a custom robot model. Ensure that the conclusion touches on each objective from the introduction, confirming if it was met. Essentially, provide a sense of closure that the research questions

have been answered.
- **7.2 Outlook – Future Work:** Offer recommendations for extending or continuing this work. Identify specific next steps, such as: *(a)* **Real-world implementation:** deploying the learned policy on the physical tendon-driven robot via ROS 2 and evaluating on real cloth, and dealing with any sim-to-real gaps (e.g. using domain randomization or fine-tuning with real data). *(b)* **Improving the policy/generalization:** exploring more advanced RL algorithms or incorporating vision-based observation so the policy can handle varying cloth shapes or even unknown cloth pieces. *(c)* **Task expansions:** e.g., moving beyond just stretching to folding cloth or handling multiple pieces, integrating higher-level planning for sequential tasks. *(d)* **Simulation enhancements:** using more accurate cloth models or including more sensors (like a depth camera for the robot to perceive the cloth state) to improve realism. Also mention any open research questions that arose: for instance, *"the adaptability of the learned policy to different fabrics (varying stiffness) remains an open question – future studies could train with a range of material properties to build a more robust policy."* Encourage that addressing these points would push the work closer to real-world application and contribute further to the field of deformable object manipulation.
- **Closing Remark:** End the thesis on a positive note, reflecting on the broader impact. For example, *"The integration of advanced simulation (Isaac Sim) with reinforcement learning demonstrated in this project exemplifies a path toward robots that can handle complex, deformable objects autonomously. As these methods mature, we edge closer to robots that can reliably perform tasks like laundry handling, recycling sorting, and more – tasks that have long been elusive due to the challenges of deformable materials."* Thank the reader if appropriate and state the final takeaway message of the thesis.

(This conclusion ensures the document is rounded off, linking back to the big picture introduced in Chapter 1. It should leave the reader with a clear understanding of what was achieved and why it matters, as well as excitement for what could follow. By keeping it within 3 pages, the writing remains focused and impactful [31] .)

Appendices (if any, not counted in main page count):

Any supplementary material such as extensive tables, hyperparameter lists, additional diagrams, or code snippets can be included in appendices. This keeps the main text lean while allowing interested readers to delve into technical details. Common appendices might include: the full URDF/MJCF of the robot, detailed network architectures, or additional results plots.

References:

A complete bibliography of sources cited, formatted according to the required style (e.g. DIN ISO 690 as per FAPS guidelines [33]). Ensure all references from literature (academic papers, textbooks, manuals like Isaac Sim documentation [34]) are listed. This thesis likely cites works in robotics, RL, and simulation, as referenced throughout the background and discussion (for instance, works on cloth manipulation, RL algorithms, and NVIDIA's technical guides).

(The overall document length for chapters 1–7 is aimed at ~70 pages, which fits the required 60–80 pages excluding front matter and appendices. The suggested page distribution – Introduction (~5), Background (~20), Gap (~3), Methodology (~15–18), Results (~15), Discussion (~5), Conclusion (~3) – is designed to meet this range. Notably, the bulk (~2/3) of the pages are devoted to Methods, Results, and Discussion, aligning with FAPS thesis recommendations [21] .)

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1 2 7 9 15 22 23 34 Robotics_Project.pdf
file://file_00000000c42071f5bababba7aaabf7f8

3 8 10 Orbit: A Unified Simulation Framework for Interactive Robot Learning Environments
https://arxiv.org/html/2301.04195v2

4 [1703.06907] Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World
https://arxiv.org/abs/1703.06907

5 25 GitHub - isaac-sim/IsaacLab: Unified framework for robot learning built on NVIDIA Isaac Sim
https://github.com/isaac-sim/IsaacLab

6 Unfolding the Literature: A Review of Robotic Cloth Manipulation
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